WEEK 15 OF 15 · STAGE 3: ADVANCED METHODS

Project presentations and course review

Your study, presented, and the whole course in one view

New R skill Rerunning the whole project from a clean session

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 **Projects and review**	

TODAY'S GOALS

By the end of today you can

1. Present a scenario study to a non-specialist audience in 15 minutes
2. Summarise each stage of the course in a few plain sentences
3. Say what IAMs can and cannot tell us, using your own project
4. Hand in code that rebuilds every figure in a fresh R session

? **Big question** Can you state the 'if' of your scenario study so clearly that a non-specialist could argue with it?

Every model in this course said "if"

En-ROADS, Hector, DICE, GCAM and the linear program all answered one kind of question: if we assume this, what follows?



What is the "if" of your project, in one sentence?

KEY IDEA

Stage 1, foundations: the building blocks

1

Climate chain

Emissions build up CO2, which traps heat and warms the planet. Warming follows cumulative emissions (weeks 1 and 2).

2

Kaya

CO2 = population × income × energy per dollar × CO2 per unit of energy (week 3).

3

Prices and time

A carbon price adds emission factor × price to costs; discounting decides how much the future counts (week 4).

4

Models and scenarios

DICE joins it all in one loop; SSPs and RCPs organise many futures (weeks 5 and 6).

Stage 2, inside GCAM: how one big model works

1

Markets

Prices adjust until supply meets demand in every region (week 7)

2

Logit choice

Market shares spread by relative cost, not winner takes all

3

Energy, land, water

Technologies, crops and bioenergy compete (weeks 8 and 9)

4

Policy

Carbon prices, caps and net-zero targets change the mix (week 10)

R grew alongside: rgcam queries, rchart and rmap figures, one change traced through gcamdata

SIDE BY SIDE

Stage 3, methods and limits: reading any model critically

Week	Big idea	One line
11	Optimisation	A cap's shadow price is a carbon price
12	Disagreement	Structure, inputs, method differ
13	Uncertainty	Ensembles show which inputs matter
14	Limits	Removal, equity, transparency, ageing

Together with stage 2, these are the skills your project uses

AN EVERYDAY PICTURE

An IAM is a flight simulator, not a crystal ball



Pilots train in simulators to learn what happens if an engine fails, not to predict their next flight. An IAM is the same: it shows what follows from a set of choices and assumptions, so we can compare options before trying them in the real world.

What it means for models

1. It explores consequences, it does not predict
2. Its answers are only as good as its assumptions
3. Comparing runs is where the insight lies

COMPARE

What IAMs can and cannot tell us

They can

- Show what follows from stated assumptions, consistently
- Compare options: carbon price, technology, timing
- Reveal trade-offs between energy, land and water
- Show which assumptions matter most

They cannot

- Predict the future or give probabilities on their own
- Decide who should pay or what is fair
- Count effects they leave out, such as many damages
- Promise that planned technologies will be delivered

SIDE BY SIDE

Your report in five parts

Part	What goes in it	Weeks
1 Question	One question a minister understands	1, 6
2 Scenario design	Reference, 2 variations, 1 sensitivity run	7 to 10
3 Results	Differences from Reference	8, 14
4 Sensitivity test	Does the main finding survive?	13
5 What it cannot say	The four-question checklist on yourself	12, 14



About 3,000 words, with R scripts that rebuild every figure

HOW IT WORKS

A fifteen-minute talk for non-specialists

1

0 to 2 min

The question and why anyone should care

2

2 to 5 min

The "if": your scenarios in one table

3

5 to 11 min

Three charts at most, each title a finding

4

11 to 13 min

The sensitivity test: does the finding survive?

5

13 to 15 min

What the model cannot tell us, and one recommendation

Imagine the audience is a farmer, a minister and your grandmother

Worked example: planning the runs

Problem Question: what does delaying a carbon price by ten years do? Each run takes about 2 hours; the team can run 2 at once.

1. Scenarios: Reference, price from 2030, price from 2040 = 3 runs
2. Sensitivity test: repeat both price runs with limited CO2 storage = 2 more
3. Total: 3 + 2 = 5 runs
4. Rounds: 5 ÷ 2 = 2.5, so 3 rounds
5. Time: 3 rounds × 2 hours = 6 hours

Answer **5 runs, about 6 hours of computing**

Quick check

A team reports 2050 CO2 of 30 Gt in its policy run. What is missing?

A Nothing, that is the result

B The difference from the Reference run

C A bigger font

D The model's source code

Quick check: answer

A team reports 2050 CO2 of 30 Gt in its policy run. What is missing?

A. Nothing, that is the result

B. The difference from the Reference run ✓

C. A bigger font

D. The model's source code

WATCH OUT

Common mix-ups on presentation day

Not quite	Better
More scenarios make a better project	A sharp question with Reference, two variations and one honest sensitivity test beats ten runs.
Show every chart the scripts made	Three to five charts, each with a title that states a finding.
The code ran last week, so it will run for the marker	Restart R and rebuild every figure from the folder before handing in.

NEW WORDS THIS WEEK

Words to add to the class glossary

Reference scenario The run without your policy or change. Every result is read as a difference from it.

Variation A run that changes one stated thing compared with the reference, such as a carbon price from 2030.

Sensitivity test Repeating key runs with one uncertain assumption changed, to see whether the main finding survives.

Presentation day checklist

Make sure your talk fits 15 minutes, your figures state findings, and your code rebuilds the whole report in a fresh R session.

4 steps

Stuck? Read the error aloud, check the slide, then ask

LAB · STEP 1 OF 4

Rebuild everything from a clean start

Restart R (Session, Restart R), then knit the report. Every figure must come from the scripts.

```
# after Session > Restart R:
rmarkdown::render("report.Rmd")
list.files("output")   # all figures rebuilt?
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Cut to three to five charts

2

Keep only charts that support your finding. Rewrite each title as a finding with units on the axis.

Rehearse to 15 minutes

3

Time a full run-through using the five-part plan. Cut detail, not the sensitivity test or the limits.

Prepare one question for another team



Use the four-question checklist from week 14 to write one fair question for the team before you.

Did it work?

You should see

- The report knits in a fresh session and rebuilds every figure
- The talk fits 15 minutes with three to five charts
- Your slides include the sensitivity test and what the model cannot tell us

Worksheet questions

1. Write the 'if' of your study in one sentence a non-specialist could argue with.
2. Name your main finding and the one assumption most likely to change it.
3. For next year's class: which lab most needed more time, and why?

RECAP

Take-home messages

1. Every IAM answers 'if we assume this, what follows?'; state your 'if' clearly
2. Read every result as a difference from Reference, and test whether it survives
3. IAMs compare options consistently; they do not predict, price fairness or promise delivery
4. A good project is small, reproducible and honest about its limits

After the course

The course ends here. Keep your project repository: it is a complete, reproducible scenario study you can show others.

Practise

Interactive course, week 15: try the project explorer and the 5-question quiz.

Readings: Your own project report and code; Revisit: Carbon Brief (2018), Q&A: How 'integrated assessment models' are used to study climate change